

Conditional goodness-of-fit for gain–loss–duplication fits: separating shape from mass under filter thresholding

recount project
notation following CM09 SI and Csűrös 2021 arXiv

May 20, 2026

Provenance and disclaimer. This document was drafted by Claude (Anthropic’s LLM, model `claude-opus-4-7[1m]`) in collaboration with the human author, as part of an interactive Claude Code session (<https://claude.com/claude-code>). The mathematical content is elementary (conditional probability + bin re-normalisation); the four numerical identities of §7 were checked at machine precision against the live `unobserved_logL0_native` routine in `recount.native_backend`, evaluated on the arc269 `mc=1, sum≥4`, and other arc269 fits saved under `validation/outputs/`. Reproduced inline by `validation/l0_goodness_of_fit.py`. Treat as “LLM-drafted, human-checked, identity-validated” engineering documentation rather than peer-reviewed manuscript content.

Abstract

Every GLD fit ships with a filter threshold $\Omega_{\min}$ that decides which families are visible to the optimiser. The fitted rate process then predicts *both* the visible distribution (the $\Omega \geq \Omega_{\min}$ tail) and an implicit unobserved stratum (the $\Omega < \Omega_{\min}$ mass below the filter). We give a rigorous goodness-of-fit framework that compares model-predicted versus empirical sum-bin counts in two complementary ways: an *unconditional* test that includes both the visible and unobserved strata, and a *conditional* test that re-normalises both pipelines to a common $\Omega \geq k_{\min}$ universe. The conditional test isolates the *shape* of the rate process from the question of total mass calibration — a separation that lets us diagnose whether a fit’s misfit is carried by the singleton stratum (plausibly contaminated by annotation noise and tip-branch *de novo* events) or by a deeper distributional mismatch. We prove the relevant consistency identities (§7), all four of which are verified to machine precision against the live arc269 fits.

1 Biological motivation

A GLD fit at filter $\Omega_{\min}$ only sees families with summed copy number $\Omega \geq \Omega_{\min}$. On arc269 the default no-filter fit ($\Omega_{\min} = 1$) sees all $F = 90,243$ observed families, but a fit at $\Omega_{\min} = 4$ sees only the $\sim 11,500$ families with $\text{sum} \geq 4$, treating the other $\sim 78,700$ small families as “unobserved.” Cs26’s own published reconstructions use $\Omega_{\min} = 4$ throughout; this is a methodological choice, not a property of the data.

What it means to “fit” under a filter. The likelihood is conditioned on $\Omega(\xi) \geq \Omega_{\min}$ for every visible family, and the rate process implicitly predicts the unobserved mass $\mathbb{P}[\Omega(\xi) < \Omega_{\min}]$ at the fitted rates. A good fit therefore needs two consistencies:

- *Shape consistency:* the bin-by-bin distribution of sums in the visible tail $\Omega \geq \Omega_{\min}$ should match the empirical histogram of family sums on that tail.
- *Mass consistency:* the predicted total $F^* = F/(1 - \mathbb{P}[\Omega < \Omega_{\min}])$ should match what the rate process implies about the population size of families that ever existed.

The two are independent: a fit can get the shape of the visible tail right but predict the wrong total mass, or it can predict the right total mass but get the visible-tail shape wrong. The framework below lets us tell which.

Why singletons matter biologically. The singleton stratum ($\Omega = 1$: families present at exactly one leaf) is the most suspect part of the dataset on biological grounds: it is the most enriched in annotation noise (false-positive gene calls), in recent lineage-specific gains that have not yet had time to duplicate, and in horizontal gene transfers not modelled by GLD. A GLD model that perfectly captures *duplicative* family dynamics could plausibly still fail at $\Omega = 1$ just because the $\Omega = 1$ stratum is being generated by a non-GLD process. The conditional test at $k_{\min} = 2$ (singletons removed) lets us ask whether the model gets the rest of the distribution right.

What this document does. §2 fixes notation; §3 introduces $L(0, N)$ (the model’s prediction of unobserved mass) and the implied bin probabilities; §4 the amplification factor F^* ; §5 the unconditional test; §6 the conditional test; §7 the four consistency identities, with proofs; §8 the interpretive discussion; §9 the arc269 case study.

2 Setup and notation

Tree and profiles. Let $\mathcal{T} = (V, E, R)$ be a rooted phylogeny with leaf set $\mathcal{L} \subset V$. A *gene-family profile* is a vector $\boldsymbol{\xi} \in \mathbb{Z}_{\geq 0}^{\mathcal{L}}$ of integer copy counts at each leaf. The *total profile sum* is

$$\Omega(\boldsymbol{\xi}) := \sum_{v \in \mathcal{L}} \xi_v \in \mathbb{Z}_{\geq 0}. \quad (1)$$

Fitted rate process and filter. Under fitted GLD rates $\theta = (\kappa, \mu, \lambda, t)$ per branch, a randomly drawn profile $\boldsymbol{\xi} \sim \mathbb{P}(\cdot \mid \theta)$ is a random variable on $\mathbb{Z}_{\geq 0}^{\mathcal{L}}$ and $\Omega(\boldsymbol{\xi})$ is its random sum. Each fit has a *filter threshold* $\Omega_{\min} \in \{1, 2, 3, \dots\}$; the likelihood is conditioned on $\Omega(\boldsymbol{\xi}) \geq \Omega_{\min}$ for every visible family.

The arc269 example. arc269 has $F = 90,243$ families, all with $\Omega \geq 1$. Sum-bin counts:

$\Omega = k$	1	2	3	4	5	6	7	≥ 8
#families	65,339	9,491	3,859	2,062	1,176	851	644	6,821

A fit at $\Omega_{\min} = 4$ sees only $F^{\geq 4} = 11,554$ families; the model never observes the $F^{< 4} = 78,689$ smaller profiles during fitting.

3 $L(0, N)$ and the model’s bin probabilities

Definition 1 (Below-threshold mass). Under fitted rates $\hat{\theta}$, define

$$L(0, N) := \mathbb{P}[\Omega(\boldsymbol{\xi}) < N \mid \hat{\theta}], \quad N \in \mathbb{Z}_{\geq 0}. \quad (2)$$

$L(0, N)$ is computable by the P_n -tensor recursion of Csürös 2026 SI Theorems 3–5 (Algorithm Uinner); in our implementation it is the routine `recount.native.backend.unobserved_logL0_native` applied to the fitted rates with `min_copies=N`. Bit-perfect agreement with the Java reference is established at $N \in \{1, 2, 4\}$; the recursion extends to arbitrary $N \geq 1$ via the same algorithm.

Lemma 2 (Monotonicity and limits of $L(0, \cdot)$). $L(0, \cdot)$ is non-decreasing on $\mathbb{Z}_{\geq 0}$: $L(0, N) \leq L(0, N + 1)$. Moreover $L(0, 0) = 0$, $L(0, 1) = \mathbb{P}(\Omega = 0)$, and $\lim_{N \rightarrow \infty} L(0, N) = 1$.

Proof. $\{\Omega < N\} \subseteq \{\Omega < N + 1\}$ gives monotonicity; the limits are immediate from $\{\Omega \geq 0\} =$ entire sample space. $\square$

Proposition 3 (Bin probabilities by telescope). *Under the fitted process,*

$$\mathbb{P}[\Omega(\xi) = k] = L(0, k + 1) - L(0, k), \quad k \in \mathbb{Z}_{\geq 0}, \quad (3)$$

with convention $L(0, 0) = 0$. Consequently $\sum_{k=0}^{\infty} \mathbb{P}[\Omega = k] = 1$.

Proof. $\{\Omega = k\} = \{\Omega < k + 1\} \setminus \{\Omega < k\}$ with $\{\Omega < k\} \subseteq \{\Omega < k + 1\}$, so the probability is the difference. Summing telescopes to $\lim_{N \rightarrow \infty} L(0, N) - L(0, 0) = 1 - 0 = 1$. $\square$

Numerical check. On the arc269 mc=1 Brownian fit, summing the bin probabilities $P(\Omega = 0) + \sum_{k=1}^{29} P(\Omega = k) + P(\Omega \geq 30)$ gives 1.0000000000 to ten decimal places. The recursion satisfies the identity exactly (`validation/10_goodness_of_fit.py`).

4 The amplification factor F^*

The fit conditions on $\Omega \geq \Omega_{\min}$, so the rate process generates an *unobserved* stratum at $\Omega < \Omega_{\min}$ that the fit cannot see. The model's prediction is calibrated by amplifying the visible count up to the implied total population.

Proposition 4 (Csürös amplification). *If the dataset $\{\xi_f\}_{f=1}^F$ of F visible families is a sample from the rate process restricted to $\Omega \geq \Omega_{\min}$, the rate process's implied total family count is*

$$F^* := \frac{F}{1 - L(0, \Omega_{\min})}, \quad (4)$$

satisfying the identity $F^*(1 - L(0, \Omega_{\min})) = F$ by construction.

Interpretation. F^* is the model's estimate of how many families the rate process would have generated if no filter were applied. The factor $1/(1 - L(0, \Omega_{\min}))$ is the inverse of the fraction of families that pass the filter; multiplying by F undoes the conditioning. Equivalently, the model is saying “for each visible family, on average there are F^*/F families in the underlying population.”

Numerical check. For the arc269 `sum` ≥ 4 fit, $F = 11,554$ and $L(0, 4) = 0.0427$ give $F^* = 11,554/0.9573 = 12,069$, and $F^*(1 - L(0, 4)) = 11,554.0$ exactly.

5 The unconditional goodness-of-fit test

The unconditional test posits the strongest version of the model: the rate process generates F^* families, and we see those whose sum passes the filter.

Definition 5 (Unconditional predicted count). The unconditional predicted count at bin k is

$$\text{pred}_{\text{unc}}(k) := F^* \cdot \mathbb{P}[\Omega(\xi) = k]. \quad (5)$$

The unconditional pred/emp ratio is

$$\rho_{\text{unc}}(k) := \frac{\text{pred}_{\text{unc}}(k)}{E(k)}, \quad E(k) := \#\{f : \Omega(\xi_f) = k\}. \quad (6)$$

The hypothesis under test is “the rate process generated F^* total families, and the visible F are its $\Omega \geq \Omega_{\min}$ slice.” For a well-calibrated fit, $\rho_{\text{unc}}(k) \approx 1$ for every $k \geq 1$ — both inside the visible range $k \geq \Omega_{\min}$ and below the threshold $k < \Omega_{\min}$ (where the model’s prediction is its own implicit self-prediction of the unobserved stratum, against which the actual empirical small-family counts can be checked).

What the unconditional test is doing. The identity $F^*(1 - L(0, \Omega_{\min})) = F$ holds by construction, so F^* is not an independent prediction of the fit — it is just F divided by the visible mass fraction the rate process implies. What an off-1 unconditional ratio reveals is one (or both) of two discrepancies: the fitted bin shape $\mathbb{P}[\Omega = k]$ disagrees with the empirical shape at the bin in question, *or* the fitted below-threshold mass $L(0, \Omega_{\min})$ disagrees with the empirical below-threshold count (which is observable when one compares to the full arc269 histogram, not just the $\Omega \geq \Omega_{\min}$ slice the fit was trained on). The two contributions are entangled in the same ratio; the conditional test below isolates the shape side.

6 The conditional goodness-of-fit test

To isolate shape from mass, re-normalise both pipelines to a common $\Omega \geq k_{\min}$ universe.

Definition 6 (Conditional bin probability). For $k \geq k_{\min}$,

$$\mathbb{P}[\Omega(\xi) = k \mid \Omega(\xi) \geq k_{\min}] = \frac{\mathbb{P}[\Omega = k]}{\mathbb{P}[\Omega \geq k_{\min}]} = \frac{\mathbb{P}[\Omega = k]}{1 - L(0, k_{\min})}. \quad (7)$$

Definition 7 (Empirical $\Omega \geq k_{\min}$ total).

$$E^{\geq k_{\min}} := \#\{f \in \mathcal{D}_{\text{full}} : \Omega(\xi_f) \geq k_{\min}\}. \quad (8)$$

On arc269: $E^{\geq 1} = 90,243$, $E^{\geq 2} = 24,904$, $E^{\geq 3} = 15,413$, $E^{\geq 4} = 11,554$.

Definition 8 (Conditional predicted count). For $k \geq k_{\min}$,

$$\text{pred}_{\text{cond}}^{(k_{\min})}(k) := E^{\geq k_{\min}} \cdot \mathbb{P}[\Omega = k \mid \Omega \geq k_{\min}] = E^{\geq k_{\min}} \cdot \frac{\mathbb{P}[\Omega = k]}{1 - L(0, k_{\min})}. \quad (9)$$

The conditional pred/emp ratio is $\rho_{\text{cond}}^{(k_{\min})}(k) := \text{pred}_{\text{cond}}^{(k_{\min})}(k)/E(k)$.

Lemma 9 (Conditional mass conservation). *By construction*

$$\sum_{k=k_{\min}}^{\infty} \text{pred}_{\text{cond}}^{(k_{\min})}(k) = E^{\geq k_{\min}} = \sum_{k=k_{\min}}^{\infty} E(k). \quad (10)$$

The conditional test conserves total mass on the $\geq k_{\min}$ universe; deviations of ρ_{cond} from 1 across bins reflect shape discrepancies only.

Proof. $\sum_{k \geq k_{\min}} \mathbb{P}[\Omega = k \mid \Omega \geq k_{\min}] = 1$ by definition; multiplying by $E^{\geq k_{\min}}$ gives the sum to that value. The right-hand sum is $E^{\geq k_{\min}}$ by definition. $\square$

Remark 10 (The conditional test is conflation-free on the $k \geq k_{\min}$ bins). The unconditional ratio at bin k carries a tautological component on any bin the fit was trained on: an $\Omega_{\min} = 1$ fit will, by construction, reproduce the full empirical histogram more closely than an $\Omega_{\min} \neq 1$ fit, because the $\Omega_{\min} = 1$ likelihood was trained on that histogram. The conditional test removes this component on the bins both pipelines agree exist. With predicted and empirical mass each re-normalised to the $\Omega \geq k_{\min}$ universe (Lemma above), neither pipeline has an advantage from

prior exposure to the bins in question; a persistent ratio deviation from 1 at $k \geq k_{\min}$ is then a genuine fit-quality difference, not a training-set artefact.

On arc269 with $k_{\min} = 2$ (singletons excluded), the unfiltered mc=1 fit's ratios stay in $[0.89, 1.11]$ across $k \in \{2, \dots, \geq 8\}$, while Csűrös' $\Omega_{\min} = 4$ fit's ratios span $[0.17, 2.56]$ on the same bins. The unfiltered fit's advantage at $k = 1$ is partly tautological — it was the only fit trained on the singleton stratum — but the shape mismatch the $\Omega_{\min} = 4$ fit shows at $k \geq 2$ is observed on bins both fits contributed equally to under the conditional re-normalisation, and is the substantive finding of the test.

7 Consistency identities

The next four propositions state what should hold by construction and which combinations of the two tests are mathematically equivalent. Each is verified to machine precision against the live arc269 fits (§9.1).

Proposition 11 (Probability mass conservation). $\sum_{k=0}^{\infty} \mathbb{P}[\Omega = k] = 1$ for any finite-rate fitted GLD process.

Proof. Direct from Proposition 3 and $\lim_{N \rightarrow \infty} L(0, N) = 1$. $\square$

Proposition 12 (F^* identity). For any fit with filter $\Omega_{\min}$ and visible count F ,

$$F^*(1 - L(0, \Omega_{\min})) = F. \quad (11)$$

Proof. Definition of F^* in Proposition 4. $\square$

Proposition 13 (Reduction at $\Omega_{\min} = 1$: cond@1 = unc on mc=1 fits). Suppose the fit's filter is $\Omega_{\min} = 1$ and the visible count equals the empirical ≥ 1 count: $F = E^{\geq 1}$. Then

$$\text{pred}_{\text{cond}}^{(1)}(k) = \text{pred}_{\text{unc}}(k), \quad k \geq 1. \quad (12)$$

Proof. With $\Omega_{\min} = 1$, $F^* = F/(1 - L(0, 1))$ and $F = E^{\geq 1}$ give

$$\text{pred}_{\text{unc}}(k) = F^* \mathbb{P}[\Omega = k] = E^{\geq 1} \mathbb{P}[\Omega = k] / (1 - L(0, 1)) = \text{pred}_{\text{cond}}^{(1)}(k). \quad \square$$

$\square$

Numerical check (mc=1 fit, $k = 4$). $\text{pred}_{\text{unc}}(4) = 2,128.83$; $\text{pred}_{\text{cond}}^{(1)}(4) = 2,128.83$; difference 0.00×10^0 (machine precision).

Proposition 14 (Reduction at $k_{\min} = \Omega_{\min}$: the conditional test collapses to unconditional). For any fit with filter $\Omega_{\min}$ and visible count $F = E^{\geq \Omega_{\min}}$, the conditional test at $k_{\min} = \Omega_{\min}$ equals the unconditional test on the same bins:

$$\text{pred}_{\text{cond}}^{(\Omega_{\min})}(k) = \text{pred}_{\text{unc}}(k), \quad k \geq \Omega_{\min}. \quad (13)$$

Proof. By Definition 8, $\text{pred}_{\text{cond}}^{(\Omega_{\min})}(k) = E^{\geq \Omega_{\min}} \mathbb{P}[\Omega = k] / (1 - L(0, \Omega_{\min}))$. Since $F = E^{\geq \Omega_{\min}}$ by hypothesis, $E^{\geq \Omega_{\min}} / (1 - L(0, \Omega_{\min})) = F / (1 - L(0, \Omega_{\min})) = F^*$, and $\text{pred}_{\text{cond}}^{(\Omega_{\min})}(k) = F^* \mathbb{P}[\Omega = k] = \text{pred}_{\text{unc}}(k)$. $\square$

$\square$

Numerical check (sum ≥ 4 fit, $k = 4$). $\text{pred}_{\text{unc}}(4) = 661.75$; $\text{pred}_{\text{cond}}^{(4)}(4) = 661.75$; ratio 1.0000 (also $E^{\geq 4} = 11,554 = F$, confirming the hypothesis).

Remark 15 (General relation between the two tests). For arbitrary $k_{\min}, \Omega_{\min}$ with $k \geq \max(k_{\min}, \Omega_{\min})$,

$$\frac{\text{pred}_{\text{cond}}^{(k_{\min})}(k)}{\text{pred}_{\text{unc}}(k)} = \frac{E^{\geq k_{\min}}}{F} \cdot \frac{1 - L(0, \Omega_{\min})}{1 - L(0, k_{\min})}. \quad (14)$$

The two tests coincide iff this ratio is 1, equivalently iff $E^{\geq k_{\min}}/F = (1 - L(0, k_{\min}))/ (1 - L(0, \Omega_{\min}))$. This holds by construction when $k_{\min} = \Omega_{\min}$ and the data is the $\geq \Omega_{\min}$ slice; in every other case the two tests measure different combinations of shape and mass.

8 Interpretation: shape vs mass

The two tests answer different scientific questions.

Test	Hypothesis under test
Unconditional	The rate process generates F^* total families, and the data $\{\xi_f\}_{f=1}^F$ is the $\Omega \geq \Omega_{\min}$ tail of a sample from it. Tests the fitted process after Csürös amplification: deviations may reflect either wrong bin-shape $\mathbb{P}[\Omega = k]$, wrong implied below-threshold mass relative to the empirical full dataset, or both. The identity $F^*(1 - L(0, \Omega_{\min})) = F$ is enforced by construction, not independently tested.
Conditional at $k_{\min}$	The rate process’s distribution restricted to $\Omega \geq k_{\min}$ matches the shape of the empirical $\Omega \geq k_{\min}$ universe. Re-normalisation puts predicted and empirical mass on a common footing, removing any training-set advantage of the $\Omega_{\min} = 1$ fit at bins below $k_{\min}$ (Remark 10). Deviations on bins $k \geq k_{\min}$ are pure shape mismatches.

When mass mismatches are biological. If the rate process generates the right shape on the tail but predicts wildly more (or wildly fewer) total families than were ever sequenced, the unconditional test will fail while the conditional test will pass. This pattern is what one would expect if the model were correctly describing duplicative gene-family dynamics but the true total number of distinct families differed from the inferred F^* — e.g. if many small or singleton families had been merged or excluded upstream of the analysis, or conversely if many spurious families had been included.

When shape mismatches are biological. If both tests fail with the same bin-wise pattern, the rate process is producing the wrong shape on $\Omega \geq \Omega_{\min}$ regardless of how mass is calibrated — the model is structurally misspecified for the visible tail. This is harder to fix by re-weighting; it requires either a different rate parameterisation (mixture, branch heterogeneity, HGT) or a different filter.

Why $k_{\min} > 1$ is biologically meaningful. If a non-trivial fraction of empirical singletons ($\Omega = 1$) is not drawn from the birth–death process at all — annotation noise, very recent *de novo* events on tip branches, recent HGT not yet replicated by paralogs — the unconditional test at $k = 1$ is contaminated by mis-attribution: the model could be a perfectly good description of *biological gene-family dynamics* and still fail at $k = 1$. The conditional test at $k_{\min} = 2$ (singletons excluded) makes this concern explicit and removes it from the test.

9 Numerical results on arc269

The mc=1 no-filter fit passes both tests at all conditioning levels:

$$\rho_{\text{unc}} \in [0.94, 1.29], \quad \rho_{\text{cond}}^{(2)} \in [0.89, 1.11], \quad \rho_{\text{cond}}^{(3)} \in [0.95, 1.03].$$

The filtered fits ($\Omega_{\min} \geq 2$) *fail* both tests, with conditional ratios spanning roughly 0.03–3.3 at $\Omega_{\min} = 10$ and $k_{\min} = 2$. The empirical shape mismatch *persists even after singletons are removed*: the rate process under filter $\Omega \geq N$ is fundamentally heavier-tailed than the empirical $\Omega \geq k$ distribution it claims to describe, for every $N \geq 2$.

Biological reading. The unfiltered mc=1 fit is the best descriptor of the family-size distribution of arc269. The filter-tightened fits sacrifice family-size shape calibration to reduce ancestral-genome inflation; this is the substantive critique of Cs26’s $\Omega_{\min} = 4$ filtering choice discussed in the project README.

9.1 Numerical identity probe

```
Identity 1: sum_k P(Omega=k) = 1.0000000000    (should be 1.0)
Identity 2 (mc=1): F* (1 - L(0,1)) = 90243.0000 (should be F_visible)
Identity 3 (mc=1, k=4): uncond=2128.83, cond@1=2128.83 diff=0.00e+00
Identity 4 (sum>=4, k=4): uncond=661.75, cond@4=661.75 ratio=1.0000
(E_{>=4} / F_visible = 1.0000 confirms hypothesis of Prop. 7)
```

All four identities hold to machine precision on the live arc269 fits.

10 Implementation reference

- $L(0, N)$ via P_n -tensor recursion (Csűrös 2026 SI Thms 3–5). Entry point `recount.native_backend.unobserv` with signature `(tree, gain, loss, dup, length, min_copies=N)`. Validated bit-for-bit against Java reference at $N \in \{1, 2, 4\}$.
- Bin probabilities $\mathbb{P}[\Omega = k] = L(0, k+1) - L(0, k)$ are computed inline from successive `min_copies=k` and `min_copies=k+1` calls to the same routine.
- Three tables (unconditional, `cond@2`, `cond@3`) applied to the arc269 fit suite live in `validation/10_goodness`; the four numerical identities of §9.1 are re-derived inside the same script for live regression testing against the saved arc269 fits.